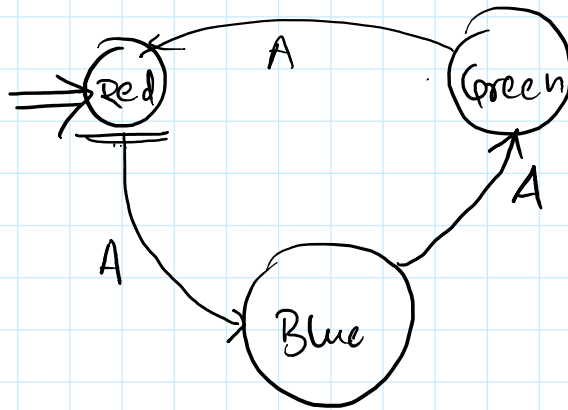


Introduction

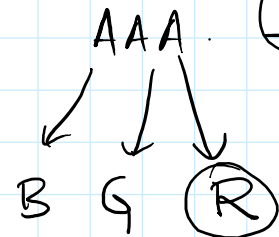
22 May 2026 14:02

States $\rightarrow$ Red, Green, ~~Not Red & Not Green~~
Blue

Operations (Transitions) $\rightarrow$ A, B.



Machine (state) $\rightarrow$ State 2



$\Rightarrow$ Green

Q> Given any multiple of 3 many A's.

$\rightarrow \emptyset$

#A's in $\emptyset = 0$.

Red

Green, Blue, Red

Finite memory $\rightarrow$ Every multiple of 3 check X.

$\{ 3, 6, 9, \dots \}$

Modulo 3 $\rightarrow \{ 0, 1, 2 \}$

$\emptyset$

2> States $\rightarrow$ ~~Red, Blue, Green~~.

2 A's in a row $\rightarrow$

$\rightarrow$ 0 A's in a row, 1 A's in a row,

2 A's in a row.

$\rightarrow$ Glow green $\rightarrow$ if the string doesn't

→ glow green → if the string doesn't contain 2 A's in a row.

[Initially it glows as green.]

∅

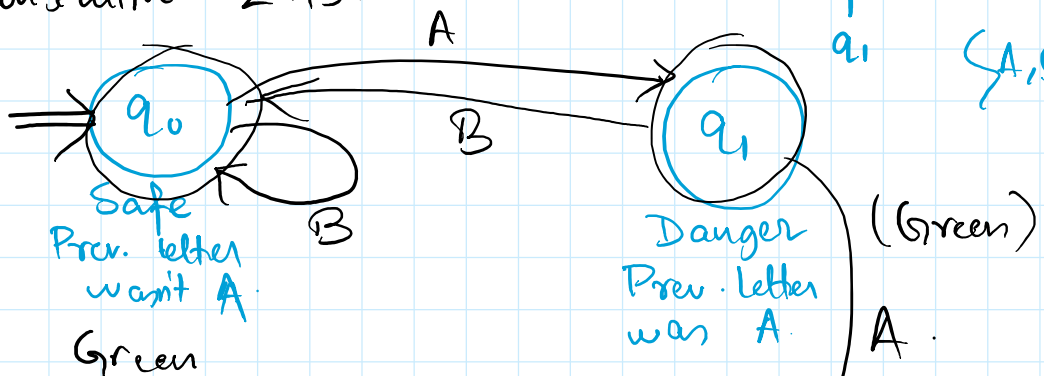
~~Prev. l~~
Prev. letter wasn't A. $\{A, B\}$

Prev. letter was A

Consecutive 2 A's.

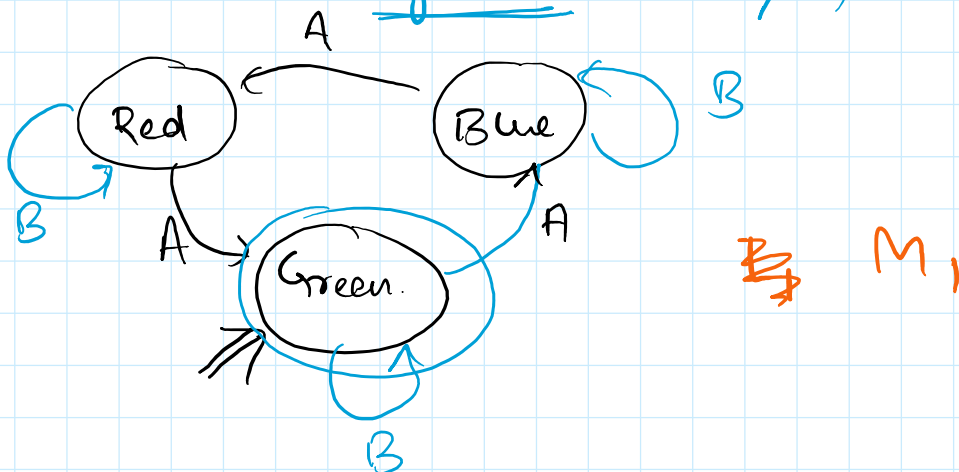
x x x x x x

q_0 A q_0 q_0 B A B A A
 $\{A, B\} \leftarrow B$



$S = \{A, B\}$
 $f: S \times S \rightarrow \text{States}$

1' Initial colour is green (∅)



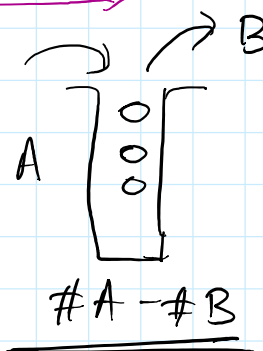
37

$$A^m(r) = A^n(r)$$

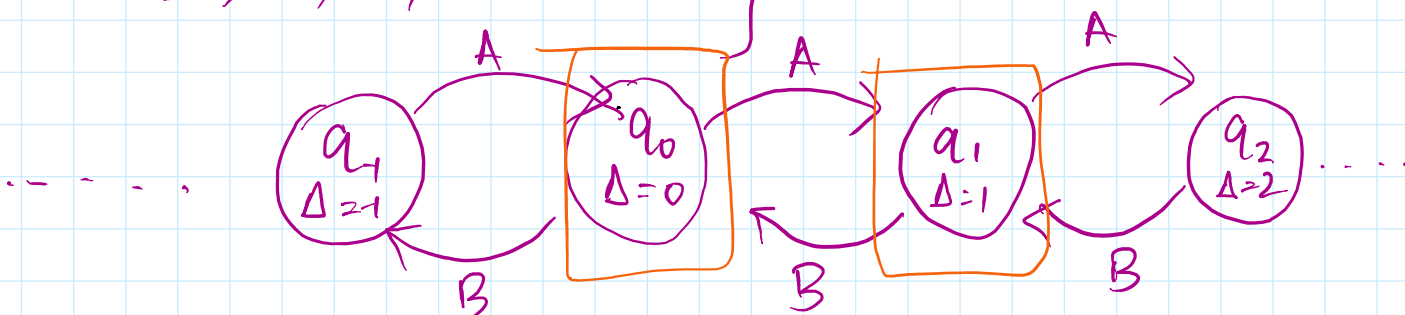
We encounter Same number of A, B's

Present Condition $(S1) \rightarrow (S2)$

$\#A - \#B = \Delta$ [Transitional]



$\{-1, -2, -1, 0, 1, 2, \dots\}$



No. of states is not finite.

[finite Automaton]

memory finite ($\Rightarrow$)

States $\rightarrow$ Current condition of the machine.

Automaton $\rightarrow$ finite memory

$\rightarrow$ Present condition (State)

$\rightarrow$ Check transitional

$\rightarrow$ Another State (based on this transitional)

Baseline $\rightarrow$ Checking entering correct password.

$\rightarrow$ Matching given string with password (saved)

$\rightarrow$ Error when we differ from our saved

$(X) \times \times \times \times$

$(X) \times \times \times \times$ saved

(xx)xxxx

(yy)yyyy saved

Compiler checking for syntax error.

Syntax match
'+' → -, -

$\left[\begin{array}{c} 'a' \\ \uparrow \end{array} + \begin{array}{c} 2 \\ \uparrow \end{array} \right]$

Expression

ϵ (AAA) → Green

M_1 accepts strings which have 0 modulo 3 A's.

Green → AA
0 modulo 3 A.

$L(M_1) = \{ w \mid \text{No. of A's (mod 3) in } w \equiv 0 \}$ → Green.

Regular / NonRegular

"Finite Automaton"

Qs → 1> Given a problem, how much memory is required to solve it?

2> Can we have 2 different machines to solve the same problem?

3> Are there problems which no machine can ever solve?

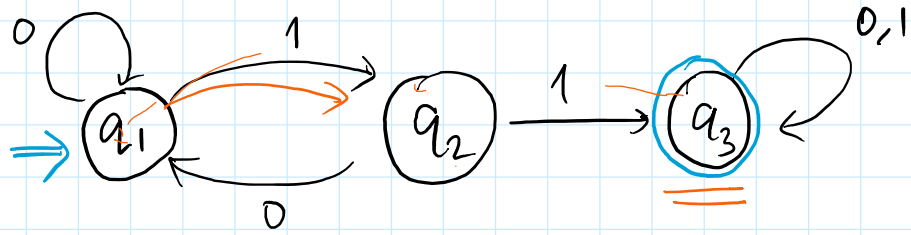
"Yes" (Based on improvement)

→ logical problems → Impossible to solve.

→ logic, Combinatorics, Algebra, linguistics
Algorithms, Complexity theory.

logic, combinatorics, algebra, linguistics
Algorithms, Complexity theory.

$Q \rightarrow M_1$



$$L(M_1) = \{ w \mid w \text{ contains substring "11"} \}$$

"~~0~~10101" $\rightarrow$ $q_1 q_1 q_2 q_1 q_2 q_1 \underline{\underline{q_2}}$

011 0 1

$q_1 q_2 \underline{\underline{q_3}}$